Sicilian–English–Italian Neural Machine Translation

A DATA PIPELINE AND REPRODUCIBLE BASELINES FROM A SMALL TRANSFORMER TO NLLB

Human Language Technologies
University of Pisa

Donato Meoli

Why is there no *good* Sicilian translator?

- For years Sicilian was **absent** from the major services. It is now included in massively multilingual models (**NLLB-200**) — but the coverage is **low-quality**: learned from non-standardised web text, not the traditional orthography. The barrier shifted from *absence* to *quality*.
- Yet Sicilian is not a "broken Italian": it is a full **Romance language** with a literary tradition older than Italian, spoken every day by millions.
- The bottleneck was never the model — it was that **nobody had assembled the parallel text**. The material exists: for 40+ years **Arba Sicula** has published a bilingual Sicilian–English literary journal.
- Wdowiak's *Tradutturi Sicilianu* (2021) was the **first** Sicilian NMT, hand-aligning that text. **This work** rebuilds it: an automated pipeline, a reproducible dataset, and modern pretrained models on top.

What is the Sicilian language?

- The **Sicilian School** of poets at the court of Frederick II created the **first literary standard in Italy** (13^{th} c.) — and *inspired Dante*, the "father of the Italian language".
- So Sicilian emerged as a **literary language before Italian**.
- It is spoken daily in Sicily, Calabria and Puglia (Italian at work, Sicilian at home) — and in diaspora communities, down to Brooklyn, NY.
- **Morphologically rich**, many local varieties, a recent (post-2017) orthographic standardisation.
- For NLP this means **genuinely low-resource**: little curated parallel text, high inflection, domain-specific (literary) language.
- Exactly the regime the methods in this talk are built for.

Roadmap

One linear story, baseline → ours:

- **data**: PDFs → parallel sentences with sentence embeddings;
- **baseline**: the Transformer, subwords, the low-resource recipe;
- **linguistic levers**: desinence-biased subwords, lemma source factors;
- **pretrained**: NLLB-200 + parameter-efficient adaptation;
- **augment**: bidirectional fine-tune, back-translation.

Contributions:

- an automated PDF→parallel-text pipeline (PyMuPDF + LaBSE), replacing the manual pdflatex+hunalign flow;
- a reproducible, provenance-tracked dataset with a held-out *literary* test set, and a BLEU/chrF harness;
- the paper's recipe ported to a Perl-free stack;
- NLLB-200 fine-tuning that **beats the published baseline** on our harder test.

Plan: data pipeline → baseline NMT → linguistic levers → pretrained & adaptation → results.

Building the parallel corpus

- The corpus bottleneck
- Extraction: PDF $\rightarrow$ language-classified pages
- Sentence alignment with LaBSE
- The unified dataset

Feng et al., *Language-agnostic BERT Sentence Embedding* (LaBSE).
Thompson & Koehn, *Vecalign* (monotonic DP alignment).

The corpus bottleneck

- For 40+ years **Arba Sicula** has published a *bilingual* literary journal — Sicilian poetry and prose with facing English translations.
- The text exists, but only as **PDFs**: facing pages, two columns, OCR-like noise, captions and front-matter. Wdowiak selected and *hand-aligned* it with `hunalign` + a Sicilian dictionary.
- **Goal**: recover that parallel text *automatically* and reproducibly, so the dataset can grow without manual alignment.

page $2k$ Sicilian	**page** $2k{+}1$ English
poem / prose	*translation*

Even/odd facing pages give a natural **language prior**; the task is to confirm pairs and align sentences within them.

Extraction — from PDF to language-classified pages

1. **Extract** clean text with **PyMuPDF** (no pdflatex round-trip).
2. **Classify** each page's language. Score a page by its share of Sicilian stop-words / diacritics; even pages lean Sicilian, odd lean English. Keep the *facing* candidate pairs (a, b).
3. **Confirm** a pair is genuinely parallel before aligning — drop covers, indices, non-parallel matter (next slide).
4. **Segment** each page into sentences (`pysbd`, per language), filtering "furniture" (all-caps headers, page numbers, captions).

A self-contained, CPU-only pipeline: `extract_pages.py` $\rightarrow$ `build_issue.py` $\rightarrow$ `build_all.py`.

Sentence alignment with LaBSE + monotonic DP

Embed every sentence with **LaBSE** (a language-agnostic dual encoder), $u_i = f(s_i^{\text{scn}})$, $v_j = f(s_j^{\text{en}})$, and build the cosine-similarity matrix

$$S_{ij} = \frac{\langle u_i,\, v_j\rangle}{\|u_i\|\,\|v_j\|}.$$

Page confirmation: keep the pair only if the mean best-match similarity $\frac{1}{2}(\overline{\max_j S_{ij}} + \overline{\max_i S_{ij}}) \geq \tau_{\text{page}}$.

Alignment: a monotonic DP over S allowing 1–1, 1–2, 2–1 moves,

$$A_{ij} = \max \begin{cases} A_{i-1,j-1} + S_{ij} \\ A_{i-1,j-2} + \frac{1}{2}(S_{i,j-1}+S_{ij}) \\ A_{i-2,j-1} + \frac{1}{2}(S_{i-1,j}+S_{ij}) \\ A_{i-1,j} - \lambda, \quad A_{i,j-1} - \lambda \end{cases}$$

keep aligned blocks with score $\geq \tau_{\text{sent}}$.

- No dictionary, no `hunalign` — just embeddings + DP.
- Thresholds $(\tau_{\text{page}}, \tau_{\text{sent}})$ **tuned against Wdowiak's hand-aligned gold** (issues AS41–42): $0.50/0.40$ recovers a corpus comparable to his manual alignment.
- Unaligned English is *not* discarded — it becomes a monolingual pool for back-translation (later).

The unified dataset

- **Arba Sicula** (our pipeline): **14,245** sentence pairs from 44 issues.
- **Napizia "Good-Sicilian-in-NLLB"** and WikiMatrix it–scn, quality- and Corsican-filtered, deduped across sources.
- A **frozen** 1k/1k valid/test held out from Arba Sicula (literary standard, *not* FLORES) — fixed once, so every later number stays comparable.

split	pairs	source
train	27,386	Arba Sicula + NLLB
valid	1,000	Arba Sicula
test	1,000	Arba Sicula (lit.)
it–scn	11,389	WikiMatrix (aux)

deterministic, deduplicated, provenance-tracked.

Preprocessing the Sicilian language

- No single orthography $\rightarrow$ *Standard Sicilian*
- ASCII-folding (drop circumflex accents)
- Uncontraction (prepositions, *aviri*, clitics)

The orthographic standard of *Arba Sicula*; rules following Wdowiak's *Tradutturi* text preprocessing. Done before any model sees the data.

Why NLLB's Sicilian is low quality

- About **one third** of the "Sicilian" NLLB collected is local **dialect**; some is actually **Corsican**, and few sentences carry a valid translation.
- Why? Romance languages are so close that a **Corsican→English** model *appears* to work on Sicilian — so mining by surface-form cues admits Corsican and dialect as "Sicilian".
- Multilingual encoders organise around **surface form**, not a shared interlingua (Verma et al., 2026) — a Corsican-looking surface form yields Corsican output.

Our evidence (LaBSE on FLORES-200):

scn→en retrieval P@1	**99%**
vs *supported* Italian	100%
Sicilian's nearest language	**Italian**
(18%, 2× the next Romance)	

Alignability tracks standardness:

	cosine	recovered
standard prose	0.76	91%
dialect poetry	0.64	80%

Sicilian sits in LaBSE's Italian/Romance region: the same proximity lets our aligner work on an officially

Standardising the Sicilian language

- Sicilian has **no single official orthography**: many local varieties, writers omit accents, several spellings per word.
- *Arba Sicula* developed a **Standard Sicilian**; we normalise every sentence to it *before* training — the data-quality step Wdowiak did **by hand**, here automated and Perl-free.
- Cuts vocabulary sparsity (decisive in low-resource) and aligns noisy web-mined text with the literary standard.

Standardisation (sample rules):

cchiù	→	chiù
io / iu / eu	→	jo
non	→	nun
'un, 'unn	→	nun
bell-	→	bedd-
pri / pir	→	pi
picchì	→	pirchì

The standard is Cipolla's, augmented for MT clarity: strict **L** on articles (*lu/la/li*), **h** on *aviri* (*haiu, havi, hannu*), **CI** for çi (*çiuri*→*ciuri*), and **R** kept before a clitic (*farlu*).

Normalisation rules: ASCII-fold + uncontraction

ASCII-fold — drop circumflex accents (writers omit them anyway):

$$\hat{a}\to a,\quad \hat{e}\to e,\quad \hat{\imath}\to i,\quad \hat{o}\to o,\quad \hat{u}\to u$$

Uncontract preposition + article (a/cu/di/pi/nta...):

ô	→	a lu	dû	→	di lu
câ	→	cu la	ntô	→	nta lu

Uncontract the verb *aviri* and clitic pronouns:

hâ {v}	→	havi a {v}			
mû	→	mi lu	tâ	→	ti la

Before: `ntô libbru dû nannu`

After: `nta lu libbru di lu nannu`

Perl-free in `sicilian_tok.py` (verified byte-identical to the Napizia tokenizer).

Ablation (NLLB, scn→en): raw 30.6 → *std* **31.0** → *full* 31.0 BLEU. Light standardisation helps a little; the aggressive "full" preprocessing does *not* beat it — a pretrained tokenizer wants un-normalised input. Orthography is a *minor* lever; the residual is data curation.

Neural MT: the baseline

- The translation objective
- The Transformer & attention
- Subword segmentation (BPE)
- The low-resource recipe

Vaswani et al., *Attention Is All You Need*.
Sennrich et al., *Neural MT of Rare Words with Subword Units*.
Sennrich & Zhang, *Revisiting Low-Resource NMT*.

The translation objective

Neural MT models translation as a **conditional language model**: an encoder reads the source $x = (x_1, \ldots, x_n)$, a decoder factorises the target left-to-right,

$$p_\theta(y \mid x) = \prod_{t=1}^{|y|} p_\theta\big(y_t \mid y_{<t},\, x\big),$$

trained by maximising the log-likelihood of the parallel corpus $\mathcal{D}$,

$$\mathcal{L}(\theta) = \sum_{(x,y)\in\mathcal{D}} \sum_{t} \log p_\theta\big(y_t \mid y_{<t}, x\big),$$

and decoded with **beam search** for $\arg\max_y p_\theta(y \mid x)$.

- Everything that follows — Transformer, subwords, NLLB — is a different parameterisation of this same p_θ.

The Transformer — attention is all you need

No recurrence: tokens interact directly through **scaled dot-product attention**. With queries Q, keys K, values V,

$$\text{Attention}(Q, K, V) = \text{softmax}\Big(\frac{QK^\top}{\sqrt{d_k}}\Big)V,$$

run in parallel as **multi-head** attention,

$$\text{MultiHead} = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\, W^O,$$

$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$. Each layer adds a position-wise FFN, residual connections and layer-norm; positions enter via positional encodings.

- **Self-attention** in the encoder and decoder; **cross-attention** lets the decoder look back at the source.
- Directly models long-range word relations — the architecture behind every model in this talk.
- Wdowiak's baseline and NLLB are both encoder–decoder Transformers; they differ in *size* and *pretraining*.

Subword segmentation — BPE

Rare and inflected words are split into **subword units**, so the model has an open vocabulary and shares statistics across word forms. **Byte-Pair Encoding**:

1. start from characters; count all adjacent symbol pairs in the corpus;
2. **merge** the most frequent pair into a new symbol; repeat for V merges.

$$(a,b)^\star = \arg\max_{(a,b)} \ \mathrm{count}(a,b) \quad \Longrightarrow \quad \text{add token } ab.$$

- Crucial for a **morphologically rich**, **low-resource** language: Sicilian inflection explodes the word vocabulary; subwords cut sparsity.
- Wdowiak's example — *Carinisi are dogs!* tokenizes to `car@@ in@@ isi are dogs !`; biasing the merges toward textbook desinences cut the vocabulary from **14,000 to 2,000** units and gave his largest BLEU gains.
- A small **joint** scn–en vocabulary (a few thousand merges) ties the two sides.

The low-resource recipe

With little data, the default Transformer **overfits**. Sennrich & Zhang (2019) prescribe a *smaller, heavily regularised* model:

- fewer layers, smaller width; small BPE vocabulary;
- high **dropout** everywhere; label smoothing; weight tying (shared source/target/output embeddings).

Our reproduction uses **Sockeye-3** (PyTorch) with exactly this geometry.

	default	ours
layers	6	**3**
embedding	512	**256**
model size	512	**256**
attention heads	8	**4**
feed-forward	2048	**1024**

$\sim$6.6M parameters; dropout 0.4; joint BPE $\approx$4k.

Linguistic levers

- Desinence-biased subwords
- Lemma source factors
- Subword regularisation (BPE-dropout)
- Stacking the gains

Sennrich & Haddow, *Linguistic Input Features Improve NMT*.
Provilkov et al., *BPE-Dropout*.

Injecting grammar — desinence-biased subwords

- Plain BPE merges by raw frequency; it ignores that Sicilian morphology lives in the **desinences** (verb/noun endings) catalogued in the textbooks (*Dieli*, *Chiù dâ Palora*).
- **Lever B**: pre-seed / bias the joint BPE so that **textbook desinences** survive as units, plus a pure-Python tokenizer that *ASCII-folds* accents and *uncontracts* Sicilian ($\hat{o} \rightarrow a\ lu$, $d\hat{u} \rightarrow di\ lu$).
- Effect: the model sees consistent morphological endings instead of scattered character merges — **+1.7 BLEU** over the raw baseline (more in Results).

The desinences come from the resources Wdowiak first assembled to *standardise* Sicilian — Dieli's *Sicilian Vocabulary*, Cipolla's *Mparamu lu sicilianu*, and the *Chiù dâ Palora* dictionary. The Napizia Perl tokenizer is re-implemented in Python (verified identical), so the whole recipe runs Perl-free.

Linguistic input features — lemma source factors

Beyond the surface token, give each *source* position extra linguistic **factors** (Sennrich & Haddow, 2016). The input embedding becomes the combination of per-feature embeddings,

$$e_j \;=\; E_{\text{word}}(w_j) \;\oplus\; E_{\text{lemma}}(\ell_j) \;\oplus\; \dots$$

(concatenation, or here a **sum** into the 256-d source embedding). We add a single factor — the **lemma** of each Sicilian token,

$$e_j \;=\; E_{\text{word}}(w_j) \;+\; E_{\text{lemma}}(\ell_j), \qquad \ell_j = \mathrm{lemma}(w_j).$$

"**lever D**". The lemma factor has a tiny vocabulary (∼900), so it *generalises* across inflected forms.

- Tells the model which surface forms share a root — exactly the information a low-resource model cannot infer from few examples.
- Out-of-vocabulary lemmas fall back to `<unk>`, keeping the factor space small.
- Our best CPU baseline: lever D reaches the lowest validation perplexity of all Sockeye variants.

Subword regularisation — BPE-dropout

- A single BPE segmentation is brittle. **BPE-dropout** (Provilkov et al., 2020) randomly drops merge operations during training, so each epoch sees a *different* segmentation of the same sentence.
- This is data augmentation in subword space: the model learns more robust subword representations and composition — valuable when data is scarce.
- Stacks on top of the desinence-biased vocabulary ("lever C" in our experiments).

Together with simply *adding more (noisier) parallel text*, these are the within-recipe knobs we ablate before turning to pretrained models.

Pretrained multilingual NMT, adapted

- NLLB-200 (Sicilian is in!)
- Parameter-efficient fine-tuning: LoRA
- Bidirectional fine-tune
- Back-translation & the Italian bridge

NLLB Team, *No Language Left Behind*.
Hu et al., *LoRA*. Sennrich et al., *Back-translation*.
Johnson et al., *Multilingual NMT*. Fan et al., *Beyond English-Centric*.

NLLB-200 — a massively multilingual Transformer

One encoder–decoder Transformer trained on **200 languages** — and crucially **Sicilian (`scn_Latn`) is supported**. The decoder is steered to the target language by **forcing the first token**,

$$p_\theta(y \mid x, \ell_{\mathbf{tgt}}), \qquad y_0 = \langle \texttt{scn\_Latn} \rangle \text{ or } \langle \texttt{eng\_Latn} \rangle.$$

It already *knows* Sicilian-adjacent Romance languages, so even **zero-shot** it translates literary Sicilian far better than a from-scratch 6.6M model.

- We use the **1.3B** variant (also tested distilled-600M).
- Knowledge transfer from 200 languages *is* the low-resource lever the small model lacks.
- Question: how to **adapt** 1.3B parameters to our 27k pairs without overfitting or a huge compute bill?

Parameter-efficient fine-tuning — LoRA

Freeze the pretrained weights $W_0 \in \mathbb{R}^{d\times k}$ and learn a **low-rank** update (Hu et al., 2021):

$$W = W_0 + \Delta W, \qquad \Delta W = \frac{\alpha}{r} BA,$$

with $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$ and rank $r \ll \min(d, k)$. The forward pass is

$$h = W_0\, x \; + \; \tfrac{\alpha}{r}\, B(Ax).$$

Only A, B are trained (∼1.3% of parameters); W_0 is untouched.

- Fits a **T4** GPU; trains in hours, not days.
- Tiny adapters can be *merged back* into W_0 at inference — no latency cost.
- Applied to the attention projections $q, k, v,$ out with $r{=}32$, $\alpha{=}64$.
- Base in fp16, adapters in fp32 for stable mixed-precision training.

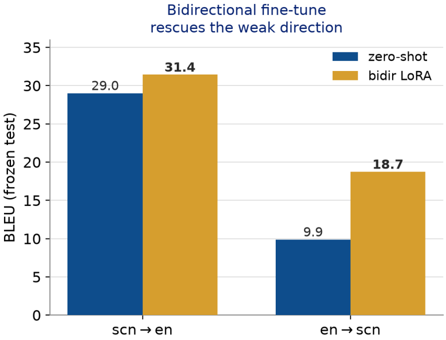
Bidirectional fine-tune — one model, both directions

Fine-tuning only scn→en leaves the reverse direction weak. Instead train **both directions at once** on the union of the data,

$$\mathcal{D}_{\leftrightarrow} = \{(x^{\mathsf{scn}}, y^{\mathsf{en}})\} \cup \{(x^{\mathsf{en}}, y^{\mathsf{scn}})\},$$

each example tagged with its forced target token.

- One adapter serves **both** directions; the shared encoder/decoder regularise each other.
- en→scn **nearly doubles** at *no cost* to scn→en — a usable reverse model, which back-translation needs next.



Back-translation — turning monolingual text into data

Given a **monolingual** target corpus $\{y\}$ and a *reverse* model $M_{t\to s}$, synthesise sources and train the forward model on real + synthetic pairs (Sennrich et al., 2016):

$$x' = M_{t\to s}(y), \qquad \mathcal{D}^{+} = \mathcal{D} \cup \{(x', y)\}.$$

The **target side** y **is real and fluent** — only the source is synthetic — so the decoder learns better target language.

Our reverse model is the bidirectional en→scn adapter; our monolingual $\{y\}$ is the **in-domain English** harvested from the *unaligned* Arba Sicula text (7,592 sentences).

- Exactly the lever that took the original system from BLEU 29.1 to 36.8.
- **Our result**: 7,498 synthetic pairs lift scn→en only $31.43 \to$ **31.6** $(+0.17)$.
- **Why so small?** His gain came from ∼**550k** back-translations; our in-domain pool is just **7.6k**. The lever is *data-limited* — not a method failure but a **data** ceiling.

Multilingual bridging — Italian as a pivot

- A single multilingual model handles many directions by prepending a **directional token** to the source, e.g. `<2it>` (Johnson et al., 2017); related languages *share* representations and enable zero-shot directions.
- Italian is far higher-resource than Sicilian and closely related. Adding **it–scn** data lets Italian **"bridge"** to Sicilian (Fan et al., 2020, *beyond English-centric*) — transferring Romance structure the English side cannot.
- We hold 11,389 it–scn pairs (+514 curated) for this auxiliary lever.

NLLB already embodies this idea at scale; we exploit it both zero-shot and via fine-tuning.

- Evaluation: BLEU & chrF
- Our models vs the baseline
- Relation to the published numbers
- Conclusion & outlook

Papineni et al., *BLEU*. Popović, *chrF*.
Post, *sacreBLEU* (comparable scoring).

Results

Evaluation — BLEU and chrF

BLEU: modified n-gram precision with a brevity penalty,

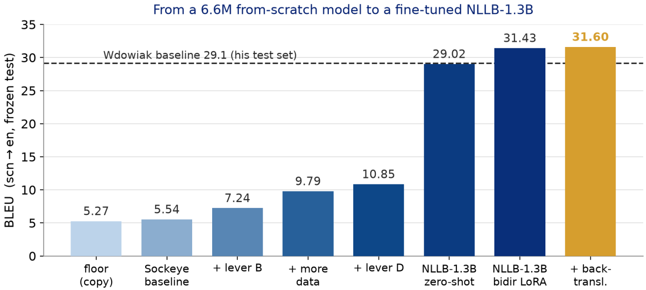
$$\mathrm{BLEU} = \underbrace{\min\left(1, e^{1-r/c}\right)}_{\text{brevity penalty}} \cdot \exp\left(\textstyle\sum_{n=1}^{4} w_n \log p_n\right).$$

chrF: character n-gram F-score, robust for morphology-rich, low-resource languages,

$$\mathrm{chrF}_\beta = (1+\beta^2)\,\frac{\mathrm{chrP}\cdot\mathrm{chrR}}{\beta^2\,\mathrm{chrP}+\mathrm{chrR}}.$$

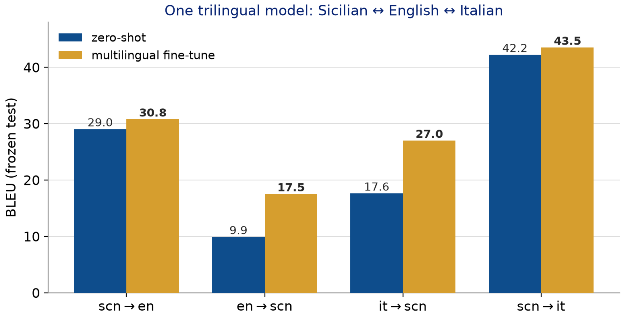
- Scored with **sacreBLEU** (`tok:13a`) for comparable numbers.
- Evaluated on the **frozen** 1k literary test set — fixed once, identical across all runs.
- Sockeye rows are tokenized-space; NLLB rows are raw text (raw floor BLEU 5.27).

Our models vs the baseline (scn→en, our test set)



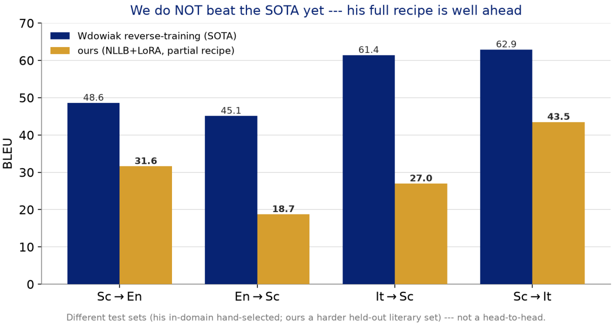
- Each Sockeye lever **stacks**: $5.54 \rightarrow 10.85$ BLEU on CPU alone (chrF $28.3 \rightarrow 35.2$).
- The pretrained model wins decisively: **+18 BLEU** zero-shot over the best from-scratch model.
- LoRA fine-tuning adds **+2.3** (to 31.43); back-translation a further $+0.17$ to **31.6** — past the published baseline on our harder test.

Adding Italian — one trilingual model



- A single multilingual LoRA adapter fine-tuned on four directions (scn↔en, it↔scn).
- **Italian is strong**: scn→it **43.5**, it→scn $17.6 \rightarrow$ **27.0** — a close, high-resource Romance neighbour (it–scn test is WikiMatrix-domain).
- Small cost to scn↔en vs the dedicated bidirectional model ($30.8/17.5$ vs $31.4/18.7$): ship the trilingual all-rounder or the specialised pair.

We do not beat the state of the art (yet)



- The SOTA is Wdowiak's **reverse-training** system; it leads on *every* direction (his Italian reaches 61–63).
- We only clear his *baseline* on Sc→en (31.4 vs 29.1), on a harder test; elsewhere we trail.
- His full recipe builds a custom pretrained model from **tens of millions** of pairs (forward + back-translation, 3 stages) — not feasible to reproduce on a single GPU.
- **But NLLB-200 already *is* that pretraining**: our run ≈ his stage-3 fine-tune. Adding back-translation + multilingual closes the *method* gap; the residual is then **data quality**.

Conclusion & outlook

- A **linear path**, fully reproduced: automated PDF→parallel-text pipeline → small high-dropout Transformer → linguistic levers (desinences, lemma factors) → **NLLB-200** with LoRA and a bidirectional fine-tune.
- On our harder literary test set we reach **31.6** (scn→en) from a 5.27 floor — above the published *baseline*, though Wdowiak's full reverse-training system is well ahead on every direction.
- **Method levers, all tested on NLLB**: multilingual Italian bridge (done), orthographic normalisation ($+0.4$), back-translation ($+0.17$, capped by our 7.6k monolingual pool vs his ∼550k). Each helps little — the gap is *not* method.
- **So the residual is data**: his hand-curated *Arba Sicula* corpus, far more back-translation, and an in-domain test set — a concrete case for **collaborating** with the author rather than competing. Also: a web app + **Telegram bot**.

Code, data pipeline and reproducible evaluation: `github.com/dmeoli/SicilianNMT`.

Acknowledgements

- **Arba Sicula**, **Gaetano Cipolla** and **Arthur Dieli** — for the bilingual journal, the vocabulary and grammar, and for sponsoring and developing Sicilian language and culture; the parallel text is used by their kind permission.
- **Eryk Wdowiak** — whose *Tradutturi Sicilianu* and "recipe for low-resource NMT" this work builds on and reproduces.
- The open-source projects this stands on: **PyMuPDF**, **LaBSE**/ sentence-transformers, **Sockeye**, **NLLB-200**, **PEFT/LoRA**, **sacreBLEU**.

Grazzi assai.

Grazzi

p'a vostra attenzioni!

github.com/dmeoli/SicilianNMT